

particles.

25.

Ans: C

When ions pass through velocity selector,

$$Eq = Bqv$$

$$20000q = 0.25qv$$

$$v = 80000 \text{ m s}^{-1}$$

When ions are at main chamber,

$$Bqv = mv^2/r$$

$$d = 2r = mv/Bq = 2(1.93 \times 10^{-25})(80000)/(0.40)(1.6 \times 10^{-19}) = 0.48 \text{ m}$$

26.

Ans: B